SWRT for nonlinear four-wave interactions in ecWAM: installation and quick guide

SWRT (sparse Webb–Resio–Tracy) computes the nonlinear four-wave interactions S_{nl} from K configurations of the WRT quadrature table, each a quadruplet and its mirror image, with one optimized coefficient per configuration. This patch adds SWRT to ecWAM as an alternative to the Discrete Interaction Approximation (DIA) in `SNONLIN`. In a 24-h global test, SWRT keeps the significant wave height within 1 % of WRT, compared with 15 % for DIA, and takes 1.6 times the time per step of DIA.

The changes, the SWRT files and this guide are on the branch `swrt` of Jialunx/ecwam, a fork of ecmwf-ifs/ecwam at commit `e1779dc0` (ecWAM 1.5.14).

1. Changes to ecWAM

The patch adds one source file and changes eight others. The default `INLSCHEME = 0` leaves ecWAM unchanged.

File	Change
`src/ecwam/snonlin_swrt.F90` (new)	`INISNONLIN_SWRT` reads the SWRT file and checks it against the spectral grid. `SNONLIN_SWRT` takes the arguments of `SNONLIN` and adds S_{nl} to `SL` and its diagonal term to `FLD`.
`src/ecwam/yowstat.F90`	Declares the switch `INLSCHEME`.
`src/ecwam/mpuserin.F90`	Reads `INLSCHEME` in namelist `NALINE`, 0 for DIA (default) and 1 for SWRT.
`src/ecwam/userin.F90`	Writes the scheme to `wam.log` and stops for other values.
`src/ecwam/initmdl.F90`	Calls `INISNONLIN_SWRT` when `INLSCHEME = 1`.
`src/ecwam/implsch.F90`	Calls `SNONLIN_SWRT` instead of `SNONLIN` in the source-term integration when `INLSCHEME = 1`.
`src/ecwam/wdfluxes.F90`	Calls `SNONLIN_SWRT` instead of `SNONLIN` for the wave-dependent fluxes at the initial time when `INLSCHEME = 1`.
`src/ecwam/CMakeLists.txt`	Adds the new file to the build.
`share/ecwam/scripts/ecwam_run_model.sh`	Passes `isnonlin`, `inlscheme` and `swrt_operator` of the yml settings to the namelist and the environment.

`SNONLIN_SWRT` computes in single precision, with the grid points of an NPROMA block as the innermost loop, and flushes subnormal numbers to zero inside this routine only. The SWRT files hold the deep-water interactions, which `ISNONLIN` scales in shallow water as it scales DIA.

As in `SNONLIN`, the diagonal term is the derivative of S_{nl} at a spectral bin with respect to the spectral density at that bin. Each quadruplet of SWRT moves action from k_3 to k_1, its two wavenumbers on the spectral grid, at a rate proportional to $T = n_1\, n_3\, (n_4 - n_2) + n_2\, n_4\, (n_3 - n_1)$, where n_i is the action density at k_i and n_2 and n_4 are interpolated. The diagonal term of the quadruplet is proportional to $\partial T/\partial n_1 = n_3\, (n_4 - n_2) - n_2\, n_4$ at k_1 and to $-\partial T/\partial n_3 = -n_1\, (n_4 - n_2) - n_2\, n_4$ at k_3, as in the WRT code Xnl (van Vledder 2006).

2. Installation

The branch can be cloned directly.

```
git clone -b swrt https://github.com/Jialunx/ecwam
```

An existing clone of ecWAM at commit `e1779dc0` takes the same changes from the patch.

```
git checkout e1779dc0aec1210ab6f5339a2582fbb7b6b18e6b
curl -LO https://raw.githubusercontent.com/Jialunx/ecwam/swrt/swrt/ecwam_swrt_e1779dc0.patch
git apply ecwam_swrt_e1779dc0.patch
```

ecWAM then builds as before and needs no new library. The SWRT files are in the folder `swrt/files` of the branch and go to any directory the model can read. The files for the other spectral grids of section 3 are in the same folder.

```
curl -LO https://raw.githubusercontent.com/Jialunx/ecwam/swrt/swrt/files/swrt_ec4x_K31L13.ww3op
curl -LO https://raw.githubusercontent.com/Jialunx/ecwam/swrt/swrt/files/wrt_table_K807.ww3op
```

SWRT file	Method	Configurations	Offsets	Use
`swrt_ec4x_K31L13.ww3op`	SWRT	31	13	runs with SWRT, selected for four times the S_{nl} cost of DIA
`wrt_table_K807.ww3op`	WRT	807	74	reference runs, all configurations with coefficient 1

3. Spectral grid

The SWRT files depend on the spectral grid only, not on the spatial grid or the time step. Each file therefore serves any spatial resolution, such as the octahedral reduced Gaussian grids O48 (about 200 km) and O1280 (about 9 km) of ecWAM. The spectral grid is set by the number of directions, the number NFRE of source-term frequencies, and the reference frequency fr1 of bin ifre1 (`fr1` and `ifre1` in the yml file). The frequency ratio of ecWAM is fixed at 1.1, and bin m has the frequency fr1 × $1.1^{m-ifre1}$. The first frequencies of the table below come from fr1 = 4.177248E-02 Hz at ifre1 = 3 (0.0345 Hz) or ifre1 = 1 (0.0418 Hz), and from fr1 = 0.055 Hz at ifre1 = 1. The standalone scripts set NFRE = 36, and `frequencies` in the yml file sets only the number of propagated frequencies.

The branch holds SWRT files for the common spectral grids below, from the configurations of ECMWF and the regression tests of ecWAM in `tests/`. Each file is selected on its own grid as the operational file is, for about four times the S_{nl} cost of DIA. In the 24-h global test of section 5.2 on the same spectral grid, SWRT keeps ε_{Hs} below 1 %, compared with 13 to 26 % for DIA.

SWRT file	Directions	NFRE	First frequency (Hz)	Ratio	Spectral grid of	ε_{Hs} SWRT / DIA (%)
`swrt_ec4x_K31L13.ww3op`	36	36	0.0345	1.1	operations, test O1280	0.7 / 15
`swrt_d24_f0345_K27L16.ww3op`	24	36	0.0345	1.1	test O320	0.3 / 17
`swrt_d24_n30_f0345_K29L14.ww3op`	24	30	0.0345	1.1	ERA5	0.3 / 20
`swrt_d24_n40_f0345_K27L16.ww3op`	24	40	0.0345	1.1	–	0.8 / 16
`swrt_d12_f0418_K29L14.ww3op`	12	36	0.0418	1.1	test O48	0.5 / 26
`swrt_d24_f0418_K27L16.ww3op`	24	36	0.0418	1.1	test O96	0.3 / 16
`swrt_d36_f055_K29L14.ww3op`	36	36	0.055	1.1	test lam_highres	0.4 / 13

A run names the file of its spectral grid in `swrt_operator` (section 4), and a change in the number of directions, NFRE, fr1 or ifre1 needs the file of the new grid. The model compares the file with its spectral grid at the start and stops with a message when they differ. The files with 30 and 40 frequencies also need NFRE = 30 or 40 in namelist NALINE, which the standalone scripts set to 36 in `ecwam_run_preproc.sh`, `ecwam_run_preset.sh` and `ecwam_run_model.sh`. Other spectral grids need a new file. The scripts in `tools/` of the SWRT code archive (doi:10.5281/zenodo.23654016) build one from Xnl in about an hour, and the code of ecWAM stays unchanged.

4. Running

Setting	Where	Value
`INLSCHEME`	namelist `NALINE`	1 for SWRT, 0 for DIA (default)
`WAM_SWRT_OPERATOR`	environment	path of the SWRT file
`ISNONLIN`	namelist `NALINE`	shallow-water scaling 0, 1 or 2, as for DIA
`WAM_SWRT_DIAG`	environment	`0` switches the diagonal term off (optional)

The standalone scripts read the same settings from three lines of the yml file.

```
physics:
    inlscheme: 1
    isnonlin: 0
    swrt_operator: /abs/path/swrt_ec4x_K31L13.ww3op
```

The log file `wam.log` names the scheme, the SWRT file, and its numbers of configurations and offsets. The model stops with a message when the variable is unset or the file is missing.

5. Test results

5.1 Synthetic spectra

The error ε of the direction-integrated S_{nl} (van Vledder 2006, Eq. 58) is computed against WRT on 300 synthetic spectra that were not used in the selection, half of them single-peaked and half bimodal. SWRT follows WRT closely, whereas the error of DIA exceeds the magnitude of S_{nl} itself.

Method	Median ε (%)	95th percentile of ε (%)
SWRT with K = 31	8	18
DIA of ecWAM	125	190

5.2 Global test

The global test extends the regression case `etopo1_oper_an_fc_048` of ecWAM to 24 h on the 36 × 29 spectral grid, from 12 UTC 31 December 2022 to 12 UTC 1 January 2023. It starts from rest with the ECMWF operational 10-m winds and sea ice, the physics of iphys = 1 and time steps of 900 s. The errors ε_{Hs} and ε_{Tz} are the relative differences from the run with WRT, summed over the hourly outputs from t = 1 h and over the sea points where H_s of WRT is at least 0.1 m.

Method	ε_{Hs} (%)	ε_{Tz} (%)	Mean H_s at 24 h (m)
WRT	–	–	1.20
SWRT with K = 31	0.7	0.8	1.19
DIA	15	9	1.35

SWRT keeps the significant wave height within 1 % of WRT at every hour and nearly everywhere, while DIA gives 15 % higher values (Figs. 1 and 2).

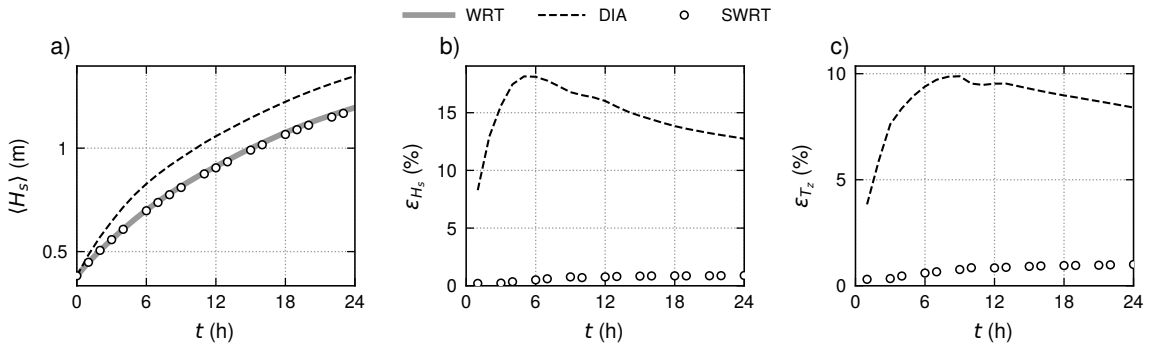


Fig. 1. Global statistics of the 24-h ecWAM test with S_{nl} from WRT (thick gray line), DIA (dashed) and SWRT with K = 31 (circles). (a) Mean H_s over the sea points; (b) ε_{Hs} and (c) ε_{Tz} at each hour.

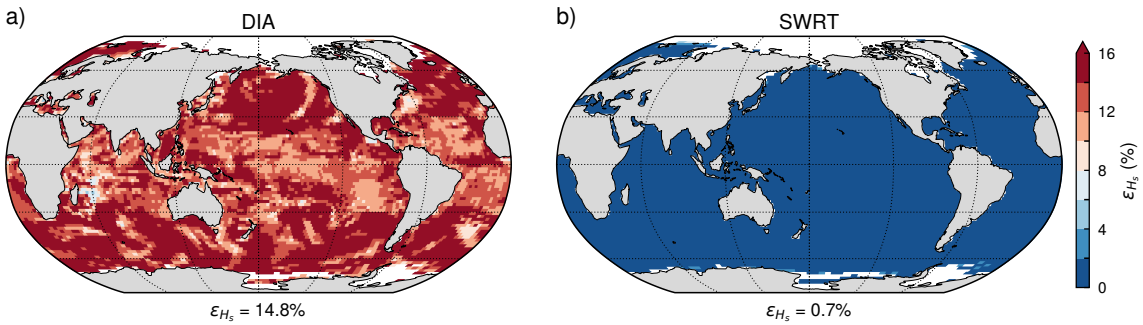


Fig. 2. ε_{Hs} at each sea point over the 24-h ecWAM test for (a) DIA and (b) SWRT with K = 31, with ε_{Hs} over all points under each map. Seas without a value are white, mostly the ice-covered polar seas.

5.3 Cost

The cost is measured with DR_HOOK in the optimized build of the 24-h runs, on one of four MPI tasks, and is given relative to DIA. WRT runs through the same vectorized routine as SWRT, and the difference in cost therefore comes from the numbers of configurations and offsets alone.

Method	S_{nl}	Time step	Run
SWRT with K = 31	4.0	1.6	1.4
WRT	68	14	11

S_{nl} takes about a fifth of a time step with DIA, half with SWRT and nearly all of it with WRT (Fig. 3).

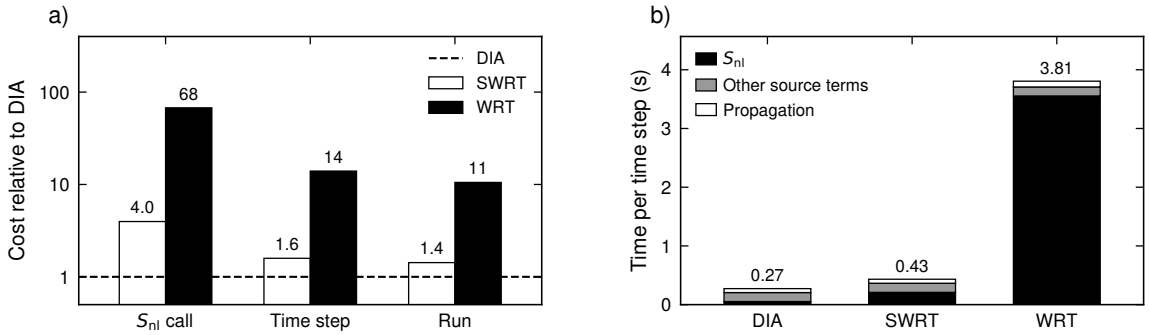


Fig. 3. Cost of S_{nl} in the 24-h ecWAM test. (a) Cost of SWRT with K = 31 (white) and WRT (black) relative to DIA (dashed line) for one S_{nl} call, one time step and the run; (b) time per time step, split into S_{nl}, the other source terms and propagation.